

Detecting Fraud in Trust Networks: A Comprehensive Approach Using Network and sentiment analysis

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Abstract

This project investigates a trust network from Bitcoin OTC, a platform where anonymous users rate each other's reputations to mitigate risks in cryptocurrency transactions. Utilizing a dataset of 5,881 users and 35,592 ratings, the study applies network analysis techniques to examine trust dynamics, identify key influencers, detect potential fraud, and uncover community structures within the network. Ratings range from -10 to +10, with 89% of ratings reflecting positive trust, leading to a network with a clustering coefficient of 0.1775 and a network density of 0.001.

Through exploratory data analysis (EDA), the project analyzed the distribution of ratings, in- and out-degree frequencies, and clustering tendencies. Temporal analysis of trust patterns showed trust trends, while centrality metrics (PageRank and degree centrality) identified influential users. Community detection using the Louvain method revealed 371 communities, highlighting groups with strong internal trust links. Fraud detection efforts focused on identifying nodes with unusual trust patterns, including users who issue extreme ratings, have non-reciprocal trust relationships, or exhibit mixed high-trust and low-trust connections.

This study demonstrates the effectiveness of network analysis in identifying suspicious behaviors and understanding trust structures in online financial networks, supporting the development of more robust trust metrics for secure transactions.

1. Introduction and Background

In decentralized networks, establishing and maintaining trust among users is a critical challenge. Unlike traditional systems, where centralized authorities manage trust and security, decentralized networks such as Bitcoin rely on peer-to-peer interactions, making them vulnerable to fraudulent behaviors and trust abuse. Users within these networks often assign ratings to each other, reflecting their perceived reliability or trustworthiness. These ratings can be leveraged to form a trust network, where nodes represent users, and edges—weighted by ratings—capture the trust or distrust levels among them.

This project aims to analyze a trust-based network in the Bitcoin ecosystem, using trust ratings between users to detect potential signs of fraudulent or suspicious behavior. The network dataset contains ratings on a scale from -10 (representing total distrust) to +10 (representing total trust) between Bitcoin users. By analyzing these trust ratings through network properties and sentiment analysis, we can identify users with unusual or risky behaviors, such as frequent extreme ratings, high-trust but non-reciprocal interactions, and mixed high and low trust connections. The key objectives of this project are:

1. To construct and analyze the Bitcoin trust network, identifying critical nodes, clusters, and patterns that reflect the underlying trust dynamics.
2. To detect indicators of fraud through techniques such as community detection, reciprocity analysis, and abrupt changes in trust ratings.
3. To explore the relationship between sentiment scores (positive vs. negative trust) and fraud indicators, assessing how negative sentiment and unusual trust behaviors may signal fraud-prone users.

This analysis not only provides insights into user behavior within a decentralized network but also highlights methods for detecting fraud in trust-based systems, which could be beneficial for developing real-time monitoring systems in decentralized environments.

2. Data Collection

This dataset represents a “who-trusts-whom” network of Bitcoin traders on the Bitcoin OTC platform, where anonymity necessitates a reputation system to mitigate transactions with potentially fraudulent or risky users. Members on this platform rate each other on a scale from -10 (indicating total distrust) to +10 (indicating full trust), making this the first available explicit weighted, signed, and directed network for research purposes.

Dataset Statistics

- **Nodes:** 5,881 unique users
- **Edges:** 35,592 trust or distrust ratings between users
- **Edge Weight Range:** -10 to +10, with steps of 1
- **Positive Edge Percentage:** 89% (indicating a majority of trust ratings)

Each record in the dataset consists of one rating event, chronologically ordered, and follows the

FORMAT:: SOURCE, TARGET, RATING, TIME

where:

- **SOURCE:** Node ID of the rater (user giving the rating)
- **TARGET:** Node ID of the ratee (user receiving the rating)
- **RATING:** Trust score from the rater to the ratee, ranging from -10 to +10
- **TIME:** Timestamp of the rating, recorded in seconds since the Epoch

This information provides a basis for analyzing user reputation dynamics, detecting patterns of trust and distrust, and exploring potential indicators of fraud within the network.

In this project, the network is a directed, weighted, and signed graph representing trust relationships among users on the Bitcoin OTC trading platform. Here, **nodes** represent individual users on the platform, and **edges** represent trust or distrust ratings given by one user to another.

- **Node:** Each node in the network corresponds to a unique user in the Bitcoin OTC platform.
- **Edge:** Each directed edge represents a rating from one user (the source) to another user (the target). The edge’s weight, which ranges from -10 to +10, indicates the degree of trust or distrust in the relationship. Positive weights reflect trust, while negative weights indicate distrust.

3. Analysis

This analysis focused on examining the trust network within the Bitcoin OTC platform to identify user trust patterns, influential nodes, community structures, and potential fraudulent behaviors. The following steps summarize the analytical approach and findings:

1.Exploratory Data Analysis (EDA):

- Distribution of Ratings:** The ratings in the network ranged from -10 to +10, with a significant skew toward positive ratings, as 89% of the interactions indicated trust rather than distrust. This distribution highlighted the general sentiment on the platform and formed the basis for trust analysis.
- Temporal Analysis of Ratings:** Ratings were resampled monthly, showing variations in trust dynamics over time, reflecting possible trends in user behaviors and trust evolution on the platform.

2.Network Metrics:

- Density and Clustering Coefficient:** The network's density was calculated to be 0.001, indicating a sparsely connected network. The clustering coefficient averaged at 0.1775, showing moderate clustering. This suggests that some users had close-knit interactions, forming trust groups within the broader network.
- Degree Distribution:** In-degree and out-degree distributions were analyzed (Average in-degree: 6.0520, Average out-degree: 6.0520), revealing that a small number of users received and issued ratings frequently, indicating influential and highly trusted individuals.

3.Community Detection:

- Louvain Method:** Community detection using the Louvain method identified 371 distinct communities, with the largest ones containing several hundred users. This result demonstrated the existence of trust-based clusters where users interact more frequently within communities than across them, forming trusted sub-networks within the platform.

4.Sentiment Analysis and Influence Metrics:

- Sentiment Ratio:** The network's overall sentiment was calculated, showing a positive-to-negative rating ratio of 0.90. This revealed a high level of trust within the network.

- Influence Scores:** PageRank was used to identify influential users within the network, spotlighting users who receive trust from many others. These highly-ranked individuals serve as trusted figures and potential influencers in the network.

5.Trust Reciprocity and Fraud Detection:

- Reciprocity Analysis:** The reciprocity ratio was 0.76, indicating that most positive trust relationships were mutual, a sign of a trustworthy network.

- Suspicious Behaviors:** Fraud detection focused on identifying users with:

- Significant trust fluctuations: Users with abrupt positive or negative changes in trust.

- Extreme Ratings: Users who frequently issued ratings at the extremes (-10 or +10), potentially showing biased behavior.

- Mixed Trust Connections: Users with both high-trust and low-trust connections, suggesting potential "trust leakage."

- Final Fraud Visualization:** A subgraph was created to visualize potentially suspicious users, categorizing them by the type of unusual behavior (e.g., trust fluctuations, non-reciprocity, extreme ratings).

Key Findings

Through these analyses, the study identified influential users, community structures, and unusual behavior patterns that may signal potential fraud. The sentiment distribution and community structures align with the presence of trust-based sub-networks, while nodes exhibiting extreme or non-reciprocal behaviors serve as early indicators of risk. The findings underscore the value of network analysis for understanding user dynamics and identifying risky interactions in anonymous online environments.

Summary and Conclusion

This project explored the trust dynamics of Bitcoin OTC, a peer-to-peer trading network where users rate each other to indicate trust or distrust. By analyzing this unique “who-trusts-whom” dataset, the study aimed to uncover patterns of user behavior, identify influential nodes, detect communities, and flag potential fraudulent activities within the network. The dataset, comprising 5,881 nodes and 35,592 edges, revealed a high prevalence of positive interactions, with 89% of ratings indicating trust, reflecting the network’s general inclination toward trustworthy relationships.

The analysis began with Exploratory Data Analysis (EDA) and temporal trends, revealing monthly variations in trust ratings and identifying influential nodes based on in-degree, out-degree, and PageRank metrics. Network metrics such as density (0.001) and clustering coefficient (0.1775) indicated a sparse yet moderately clustered network, implying some closely connected trust clusters. Community detection using the Louvain method identified 371 distinct communities, highlighting groups where users preferentially interact, forming trust-based clusters.

Sentiment analysis underscored the network’s overall positive nature, with a sentiment ratio of 0.90. Further examination of reciprocity revealed a high level of mutual trust, with a reciprocity ratio of 0.76, indicating that most positive ratings were reciprocated. Potential fraud indicators were identified by analyzing users with extreme ratings, mixed trust connections, and significant changes in trust scores. Visualizations of these identified nodes provided a clear representation of users with unusual behaviors that may warrant further investigation.

In conclusion, the project demonstrated the effectiveness of network analysis in understanding user trust dynamics within a digital platform. By examining trust patterns, influential users, and potential risk factors, this study provided insights into the social structure and reliability of the Bitcoin OTC network. The findings suggest that sentiment distribution and reciprocity are useful metrics for identifying stable relationships, while community structures and unusual behaviors can serve as early indicators of potential fraud. This analysis highlights the value of network techniques for improving the safety and integrity of anonymous, peer-to-peer trading platforms.